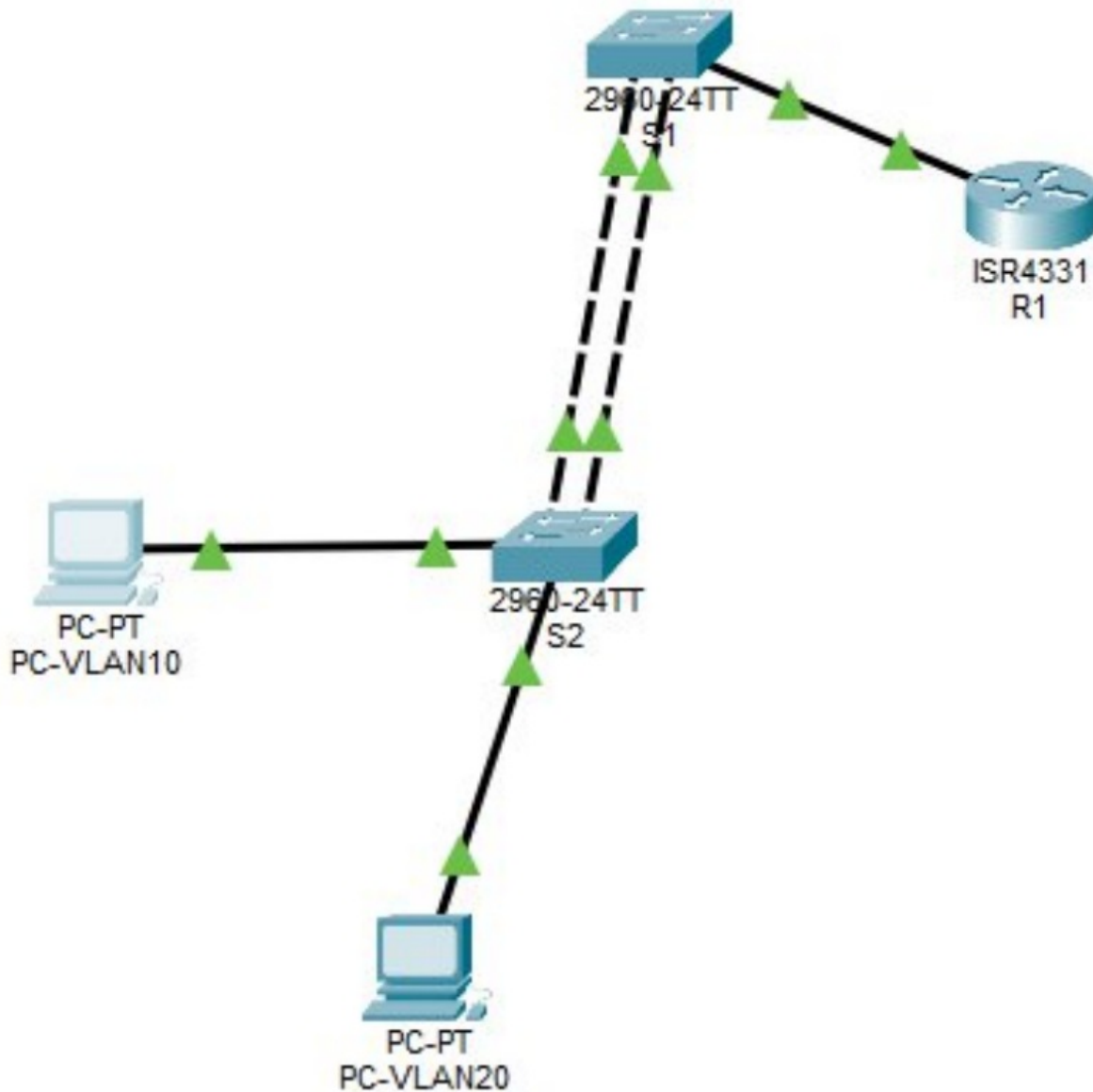


Inter-VLAN Routing (RoAS) & EtherChannel LACP

This guide covers the key theoretical concepts and full configuration procedures for deploying a redundant, segmented network architecture on Cisco IOS equipment. This scenario is representative of CCNA performance-based questions (PBQs).



Part 1 — Core Concepts

This lab combines three pillars of Cisco Layer 2/Layer 3 network infrastructure:

1. Inter-VLAN Routing (Router-on-a-Stick / RoAS)

By default, VLANs isolate traffic at Layer 2 (Data Link). For two different VLANs to communicate (e.g. Finance and Marketing), a Layer 3 (Network) device is required to route packets between them. The **Router-on-a-Stick** method uses a single physical link between a switch and a router (configured as a trunk). The router's physical interface is split into logical sub-interfaces, each assigned to a specific VLAN, encapsulating frames per the IEEE 802.1Q (dot1q) standard and holding that subnet's default gateway IP address.

2. Link Aggregation (EtherChannel via LACP)

EtherChannel bundles multiple redundant physical links (here, two cables between the core switch S1 and the access switch S2) into a single logical link called a Port-Channel. This provides two major benefits: increased aggregate bandwidth and fault tolerance (if one physical cable is cut, traffic instantly fails over to the other with no interruption). The protocol used here is **LACP** (Link Aggregation Control Protocol), an open standard (IEEE 802.3ad) that dynamically negotiates bundle formation via `active` mode.

3. Device Access Security (Baseline Hardening)

Network security starts with locking down management access to devices (control plane). This lab implements two standard protection levels:

- **Privileged secret password** (`enable secret`): prevents unauthorized users from reaching global configuration mode from user EXEC mode.
- **Console password** (`line console 0`): secures the physical serial port (RJ45/USB) used for direct local configuration on the device.

Part 2 — Topology & Addressing

1. Physical Connections

- R1 (GigabitEthernet 0/0/0) —> S1 (GigabitEthernet 0/1) *[straight-through cable]*
- S1 (FastEthernet 0/23) —> S2 (FastEthernet 0/23) *[crossover cable — EtherChannel bundle]*
- S1 (FastEthernet 0/24) —> S2 (FastEthernet 0/24) *[crossover cable — EtherChannel bundle]*
- S2 (FastEthernet 0/10) —> PC-VLAN10 *[straight-through cable]*
- S2 (FastEthernet 0/20) —> PC-VLAN20 *[straight-through cable]*

2. IP Segmentation & VLANs

- **VLAN 10 (Finance)**: subnet `192.168.10.0/24` — gateway (R1): `192.168.10.254`
- **VLAN 20 (Marketing)**: subnet `192.168.20.0/24` — gateway (R1): `192.168.20.254`

Part 3 — CLI Configuration Scripts

1. Router R1 Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname R1

R1(config)# enable secret PasswordPrivilege123
R1(config)# line console 0
R1(config-line)# password PasswordConsole123
R1(config-line)# login
R1(config-line)# exit

R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface gigabitEthernet 0/0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.254 255.255.255.0
R1(config-subif)# exit

R1(config)# interface gigabitEthernet 0/0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.254 255.255.255.0
R1(config-subif)# exit
```

2. Core Switch S1 Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1

S1(config)# vlan 10
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# exit

S1(config)# interface gigabitEthernet 0/1
S1(config-if)# switchport mode trunk
S1(config-if)# exit

S1(config)# interface range fastEthernet 0/23 - 24
S1(config-if-range)# channel-group 1 mode active
S1(config-if-range)# exit

S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# exit
```

3. Access Switch S2 Configuration

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S2

S2(config)# vlan 10
S2(config-vlan)# exit
S2(config)# vlan 20
S2(config-vlan)# exit

S2(config)# interface range fastEthernet 0/23 - 24
S2(config-if-range)# channel-group 1 mode active
S2(config-if-range)# exit

S2(config)# interface port-channel 1
S2(config-if)# switchport mode trunk
S2(config-if)# exit

S2(config)# interface fastEthernet 0/10
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
S2(config-if)# exit

S2(config)# interface fastEthernet 0/20
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 20
S2(config-if)# exit
```

Part 4 — Validation & Troubleshooting

To confirm the setup is correct, configure the PCs with their matching static IP addresses (e.g. `192.168.10.1` for PC-VLAN10) and initiate a ping to the other network. Use these three key CCNA commands to validate network state:

Verify the LACP EtherChannel bundle:

```
S1# show etherchannel summary
```

In the port flags, the Port-Channel should show `(SU)` (In Use / Layer 2) and the physical FastEthernet interfaces should show `(P)` (Bundled in port-channel).

Verify trunks:

```
S2# show interfaces trunk
```

Confirm that Port-channel 1 appears with `trunking` status and that VLANs 10 and 20 are explicitly listed as allowed and active.

📌 **Exam reminder:** don't forget to save the configuration on every device with the official command required during the exam: `copy running-config startup-config`.